

Programme: Diploma in Cloud Computing and Big Data	
Course Code: 5273B	Course Title: Advanced Data Structures
Semester: 5	Credits: 4
Course Category: Programme Elective Course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To impart concepts of graph and tree data structures.
- To impart concepts Maps, Hash Tables and Set data structures.
- To provide ideas of text processing and associated algorithms.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic concepts of Data Structures and Algorithms	4272	Data Structures and Algorithms	4
Python Programming Concepts	3279	Programming in Python Lab	3
Application of data structures	4278	Data Structures and Algorithms Lab	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate graph algorithms for transitive closure, topological sorting, shortest path and minimum spanning tree.	16	Applying
CO2	Illustrate AVL trees and their operations.	13	Understanding

CO3	Explain Maps, Hash Tables and Set data structures	14	Understanding
CO4	Summarize text processing and associated algorithms.	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate graph algorithms for transitive closure, topological sorting, shortest path and minimum spanning tree.		
M1.01	Demonstrate Floyd Warshall algorithm to find transitive closure of a directed graph	3	Understanding
M1.02	Explain directed acyclic graphs and topological sorting algorithm	3	Understanding
M1.03	Illustrate Dijkstra's shortest path algorithm	4	Applying
M1.04	Use Prim-Jarnik Algorithm to find Minimum Spanning Tree	3	Applying
M1.05	Find the Minimum Spanning Tree of a graph using Kruskal's Algorithm.	3	Applying

Contents: Transitive Closure – Definition, Floyd Warshall Algorithm, Directed Acyclic Graphs – Definition, Applications, Topological Ordering algorithm, Weighted Graph, Shortest Path - Dijkstra’s Algorithm, Spanning Tree, Minimum Spanning Tree - Prim-Jarnik Algorithm - Kruskal’s Algorithm.			
CO2	Illustrate AVL trees and their operations.		
M2.01	Describe balanced search trees	2	Understanding
M2.02	Illustrate the rotation process for tree balancing	3	Understanding
M2.03	Outline the property of AVL trees.	3	Understanding
M2.04	Explain update operations on AVL trees.	5	Understanding
	Series Test – I	1	
Contents: Balanced Search Trees, tree rotation and restructuring, AVL Trees – Definition, Update Operations			
CO3	Explain Maps, Hash Tables and Set data structures		
M3.01	Describe Map Abstract Data Type (ADT) and its operations.	2	Understanding
M3.02	Illustrate the structure of hash table	3	Understanding
M3.03	Classify hash codes and functions	3	Understanding
M3.04	Summarize collision handling approaches	3	Understanding
M3.05	Explain set ADT and its operations	3	Understanding
Contents: Sets, Multisets and Multimaps, The Map ADT, Hash Tables, Hash codes and Functions, Collision-Handling Schemes, The Set ADT - operations on sets			
CO4	Summarize text processing and associated algorithms.		
M4.01	Describe the importance of text processing.	2	Understanding

M4.02	Outline the brute-force approach to pattern matching.	3	Understanding
M4.03	Explain text compression using Huffman Coding Algorithm	3	Understanding
M4.04	Describe Tries	2	Understanding
M4.05	Summarize categories of tries	3	Understanding
M4.06	Describe Search Engine Indexing	2	Understanding
	Series Test – II	1	

Contents:

Abundance of digitized text, Pattern-Matching algorithms - Brute Force approach, Text Compression – Huffman code, the Huffman Coding Algorithm, Tries, Standard Tries, Compressed Tries, Suffix Tries, Search Engine Indexing - Inverted Files

Text / Reference

T/R	Book Title/Author
T1	Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, Data Structures and Algorithms in Python (An Indian Adaptation), Wiley; 1st edition (1 July 2021)
R1	Bradley N Miller, Problem Solving with Algorithms and Data Structures Using Python, Franklin Beedle & Assoc; Second edition (1 January 2013)
R2	Kent D. Lee, Steve Hubbard, Data Structures and Algorithms with Python, Springer, 2015

Online Resources

Sl.No	Website Link
1	https://www.edureka.co/blog/data-structures-in-python/
2	https://realpython.com/python-data-structures/
3	https://www.programiz.com/dsa